

Erdős #153: MEAN-SQUARE GAP DIVERGENCE FOR ASYMPTOTICALLY MAXIMUM SIDON SETS

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For a finite Sidon set A with $A + A = \{s_1 < \dots < s_t\}$, write $Q(A) := \frac{1}{t} \sum_{i=1}^{t-1} (s_{i+1} - s_i)^2$. Erdős Problem #153 [1] asks whether $Q(A) \rightarrow \infty$ for every sequence of finite Sidon sets with $|A| \rightarrow \infty$.

Remark 1 (Erratum). An earlier version of this note claimed a full resolution of #153 using a shifted-intersection bound $|(A + A) \cap (A + A + d)| \leq 2n^{3/2}$ attributed to [3] via Bloom [4]. Bloom has since retracted that bound [?]; the correct estimate is $r_{A+A}(d) \leq |A + A|/2 + O(|A + A|^{3/4})$, which does not suffice for the argument. The present note replaces the retracted proof with a partial result covering the asymptotically maximum case.

We prove the following:

Theorem 2. *Let (A_n) be finite Sidon sets with $|A_n| = n \rightarrow \infty$, and suppose (after translation) that $A_n \subset [0, D_n]$ with $0, D_n \in A_n$ and*

$$D_n = (1 + o(1)) n^2.$$

Then $Q(A_n) \rightarrow \infty$.

The hypothesis $D_n = (1 + o(1)) n^2$ says that A_n is asymptotically maximum: a Sidon set of size n in $[0, D]$ requires $D \geq (1 - o(1)) n^2$ by the Erdős–Turán bound, and the Bose–Chowla construction [?] achieves $D = (1 + o(1)) n^2$.

Remark 3. The complementary regime $D_n/n^2 \rightarrow \infty$ is elementary: Cauchy–Schwarz gives $Q(A) \geq 4D^2/t^2$, and $t = n(n+1)/2 \ll D$ forces $Q \rightarrow \infty$. This was observed by Dutta [?] on the Erdős Problems #153 forum. The intermediate regime $D_n \sim C n^2$ for a fixed constant $C > 1$ remains open.

Proof of Theorem 3. Let $A = A_n$, $|A| = n$, $D = \text{diam}(A) = (1 + o(1)) n^2$. Since A is Sidon,

$$t = |A + A| = \binom{n+1}{2} \sim \frac{n^2}{2}.$$

Step 1: sparse intervals force square-gap mass. Let $T = A + A = \{s_1 < \dots < s_t\}$. If an interval I has length L and contains $m = |T \cap I|$ points of T , then those m points partition I into at most $m + 1$ subintervals. By Cauchy–Schwarz,

$$(4) \quad \sum_{\text{gaps crossing } I} (s_{i+1} - s_i)^2 \geq \frac{L^2}{m+1}.$$

Step 2: Pikhurko uniformity gives edge distribution. Because A is an asymptotically maximum Sidon set in $[0, D]$, Pikhurko’s uniformity lemma [?, Lemma 10] gives, for every fixed $\alpha \in (0, 1)$,

$$(5) \quad |A \cap [0, \alpha D]| = (\alpha + o(1)) n.$$

Step 3: disjoint dyadic edge bands. Fix an integer $K \geq 1$. Define $\delta_j = 4^{-j}$ for $j = 1, \dots, K$, and take the lower-edge bands

$$I_j = [2\delta_j D, 4\delta_j D].$$

These are pairwise disjoint since $4\delta_{j+1}D = \delta_j D < 2\delta_j D$. Each has length $|I_j| = 2\delta_j D$.

Step 4: bound sums in each edge band. Every element of $(A + A) \cap I_j$ is a sum $a + b$ with $a + b \leq 4\delta_j D$, hence $a, b \leq 4\delta_j D$ (since $a, b \geq 0$). Therefore

$$|(A + A) \cap I_j| \leq \binom{|A \cap [0, 4\delta_j D]| + 1}{2}.$$

By (??), $|A \cap [0, 4\delta_j D]| = (4\delta_j + o(1))n$ for each fixed j , so

$$(6) \quad |(A + A) \cap I_j| \leq (8\delta_j^2 + o(1))n^2.$$

Step 5: each band contributes $\gg t$. By (??), (??), and $|I_j| = 2\delta_j D$,

$$\sum_{\text{gaps crossing } I_j} (s_{i+1} - s_i)^2 \geq \frac{(2\delta_j D)^2}{(8\delta_j^2 + o(1))n^2 + 1} = \frac{4\delta_j^2 D^2}{(8\delta_j^2 + o(1))n^2 + 1}.$$

Since $D = (1 + o(1))n^2$, for each fixed j and n sufficiently large this is

$$\geq \frac{(1 - o(1))n^2}{2} \gg t.$$

Step 6: sum over K disjoint bands. The intervals $I_1, \dots, I_K$ are disjoint, so the gap contributions sum without double-counting:

$$\sum_{i < t} (s_{i+1} - s_i)^2 \geq \sum_{j=1}^K \frac{(2\delta_j D)^2}{|(A + A) \cap I_j| + 1} \gg Kt.$$

Dividing by t ,

$$Q(A) \gg K.$$

This holds for every fixed K once n is sufficiently large (depending on K). Therefore $Q(A_n) \rightarrow \infty$. $\square$

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